IB 514: Scientific Writing

Mark Novak

June 1, 2026

Table of Contents I

Table of Contents II

Table of Contents III

Overview I

Introduction

Overview

Overview I

Course Goals

Primary Goal

To develop more effective and efficient writing skills.

Secondary Goal

To make significant progress on your manuscript.

What you'll need

"You must be ready to start writing a scientific manuscript the first week of the term. What does "be ready" mean? That's for you and your advisor(s) to decide.

That said, being ready does not mean being 100% done with your analyses or having all your final figures drafted. However, you do need to be far enough along to know what your paper's main story arc and primary message will be."

Core Principles

The 90-90 Rule

Doing the research is the first 90% of any project; writing it up is the other 90%.[1]

1. Being a scientist is being a writer
2. Productivity is designed, not discovered
3. Structure supercedes prose
4. Separate drafting from revising
5. Write for your readers
6. High-quality writing emerges from high-quality workflows
7. Literature is managed systematically
8. Collaboration is communication
9. Open science and reproducibility beget efficiency

[1] misquote of Tom Cargill of Bell Labs; see also Hofstadter's law.

Overview I

1 Introduction

- Philosophy
- Overview
  - Website & Materials
  - Questions?
  - Who are we?
  - Accountabilibuddies
- My hope

Schedule, Readings & Materials

`https://github.com/marknovak/ScientificWriting/`

What have I forgotten to tell you?

Who are we?

- Preferred name
- Department
- What you study
- Fun fact about yourself
- Why you're here

Accountabilibuddies

- Weekly check-in
  - *Goals/deadlines, accomplishments, sticking-points*
  - *First 5-10 min. of every Wednesday*
- Randomized pairing
  - *re-randomized week 6 (if desired)*

Overview I

We're all learning...

Be kind to yourself.
Support each other.

Integrity

Doing the right thing, even when no one is watching.

Let's get started!

`.../classes/ProjectProposal`

Overview I

2 Habits

- Daily Writing Routines
- Time Management & Deadlines
- Early Drafting Practices
- Writer's Block
- Meta-Cognition
- Assignment

Habits

Session Objectives

- Build sustainable daily writing routines
- Design protected writing time
- Structure deadlines strategically
- Improve drafting and workflow practices
- Diagnose and intervene on writer's block
- Develop meta-cognitive awareness

Overview I

2 Habits

- Daily Writing Routines
- Time Management & Deadlines
- Early Drafting Practices
- Writer's Block
- Meta-Cognition
- Assignment

Daily Writing as a System

Core Principle: Writing is a repeatable cognitive practice.

- Frequency > intensity
- Define task before session begins
- Protect high-cognition hours

Some Models of Daily Writing

1. **Consistency Model**
   - 30–60 min daily
   - Same time, same location
2. **Sprint Model**
   - 25-min focused intervals
   - 2–3 per day
3. **Cognitive Load Model**
   - Morning: argument development, drafting new sections
   - Midday: data checks, figure refinement
   - Late day: formatting, references, minor edits
4. **Paragraph Commitment Model**
   - 1 paragraph/day
   - Stop after completion

Concrete Output and Task Decomposition

- 1 paragraph/day $\rightarrow$ 5 per week
- 300 words/day $\rightarrow$ 6,000 per month
- Draft Results before Introduction

Task Decomposition

- Instead of “Revise Discussion”

  $\rightarrow$ Cut 300 words

  $\rightarrow$ Clarify paragraph 3 claim

  $\rightarrow$ Add 2 citations on stability theory

Overview I

Protected Writing Time

Treat it as a meeting with yourself.

Protected time is:

- Scheduled
- Visible
- Non-negotiable

Implementation Strategies

- Block 60–90 min during peak cognitive hours
- Close email + messaging platforms
- Define task in calendar description
- Begin immediately (no warm-up browsing)
- Start with 2–3 blocks/week; increase after consistency

Major vs. Minor Deadlines

Nested Planning Structure

Major deadlines determine direction.
Minor deadlines determine daily behavior.

Major Deadlines

- Journal submission
- Thesis chapter defense
- Circulate full draft to co-authors

Minor Deadlines

- Draft study system section of Methods by April 1
- Figure 2 drafted by Tuesday
- Circulate Introduction in 10 days
- Draft Abstract by Friday

Examples of Minor and Micro Deadlines

Examples within a 4-week revision window:

- Day 3: Rewrite Results subsection 1
- Day 5: Reduce Discussion by 400 words
- Week 2: Complete robustness checks
- Week 3: Internal logic review of Introduction
- Week 4: Format references

Micro-Deadlines

- Clarify claim of paragraph 4 (30 min)
- Verify statistical assumption 2
- Insert 3 missing citations
- Rewrite topic sentences in Results

Minor goals should be specific, time-bound, and measurable.

Cognitive Energy Management

Align task to cognitive state

High-energy tasks:

- Argument construction
- Structural revision

Medium-energy tasks:

- Sentence rewriting

Low-energy tasks:

- Formatting
- Reference checks

Overview I

Draft Before You Feel Ready

Write imperfect prose intentionally

"Write drunk, edit sober." – Hemingway

Separate drafting from editing.

- Draft full section quickly
- Revise next day

Reverse Outline

Extracting the logical structure after drafting.

Process

- Reduce each paragraph to one bullet:
  "What claim does this paragraph make?"
- Examine logical order.
- If bullets do not form a coherent argument, restructure.

Overview I

Diagnosing Writer's Block: Underlying Causes

Writer's block is usually a symptom, not a primary problem.

1. **Cognitive Causes**
   - Argument is not fully specified
   - Paragraph lacks a clear claim
   - Logical steps between ideas are missing
   - Methods or assumptions not fully understood
2. **Structural Causes**
   - No outline or unstable outline
   - Section scope too broad
   - Unclear transition between sections

Diagnosing Writer’s Block: Underlying Causes

3. **Process Causes**
   - Editing while drafting
   - Task too large (“Revise paper”)
   - No defined next action
4. **Psychological Causes**
   - Perfectionism
   - Fear of evaluation (advisor, reviewers)
   - Avoidance of cognitively demanding work

Writer’s Block

A signal that something specific needs clarification.

Interventions: Matching Strategy to Cause

If the argument is unclear:

- Write the paragraph's claim as a single sentence.
- Ask: What must be true for this claim to hold?
- Sketch logic on paper before drafting prose.

If the task is too large:

- Reduce to a 15-minute goal.
- Replace "Revise Discussion" with "Rewrite paragraph 2 for clarity."

If perfectionism is interfering:

- Draft in bullet points.
- Use timed freewriting (10 minutes, no deletion).
- Write a deliberately "bad" version.

Interventions: Matching Strategy to Cause

If fatigue is the issue:

- Switch to a lower-cognitive task (references, figure labels).
- Stop and resume during peak cognition.

If stuck mid-paragraph:

- Insert bracketed placeholders: [CITE], [EXPLAIN MECHANISM], [CHECK RESULT]

Momentum

Momentum restores clarity more reliably than rumination.

Overview I

Meta-Cognitive Writing

Strong writers monitor:

- When they write best
- How long tasks take
- What triggers avoidance
- What improves clarity

Writing Log

- Date, Duration
- Task completed
- Challenges encountered & insight gained

Example Log Entry:

- 8:00–8:45 AM, Drafted 2 Results paragraphs
- Stalled at interpretation step
- Insight: need clearer model summary before Discussion

Overview I

Assignment: (Re)Design Your Writing System

Write down:

- 2-3 protected writing blocks per week
- 2 minor deadlines for this week
- 1 drafting experiment to try (e.g., reverse outline, freewriting)

Writing productivity is designed, not discovered.

Overview I

Workflows

Framing the Session: Why Workflows Matter

Central Premise: High-quality writing emerges from high-quality systems, not last-minute effort.

Three Guiding Questions

1. How can I reduce friction between ideas and written text?
2. How can I prevent structural chaos as projects grow?
3. How can I ensure my workflow supports collaboration and revision?

Session Objectives

- Design a scalable project structure
- Identify points where inefficiency may emerge
- Evaluate writing tools strategically

Key Insight: Workflow design is a methodological decision

It shapes clarity, reproducibility, and speed.

Overview I

Project Structure

Recommended Folder Architecture

- `/notes` – reading notes, idea sketches, meeting notes
- `/data`
  - `/raw` (read-only)
  - `/processed`
- `/code` – scripts, functions, notebooks
- `/results` – exploratory outputs, intermediary results
- `/tables` - publication-ready tables[2]
- `/figures` – exported publication-ready figures
- `/manuscript` – bibliography, draft(s and versions[3])
  - `/biblio` – .bib files, reference management exports
  - `/versions` – draft versions, co-author comments
  - `/submitted` – submitted files, cover letter, reviews, review responses

[2] Exported! (e.g., `stargazer`, `kableExtra`, `flextable` R packages)
[3] If not using version control software

Project Structure

Best Practices

- Explanatory README file in root directory
- Separate raw and cleaned/processed data
- Separate exploratory from finalized results (figs and tables)
- Consistent naming conventions

File Naming Conventions

- Consistent ordering:
  - `survey_raw_2025.csv`
  - `survey_cleaned_2025.csv`

  Consistent capitalization:
  - `Upperstart` vs. `lowercase`
  - `CamelCase` vs. `start_end`
- Use descriptive, machine-readable names:
  - Avoid: `finalresults_real_final2.csv`
  - Prefer: `results/regression_modelA.csv`

Version Control

Why Version Control?

- Prevents loss of work
- Eases identification and fixing of problems/bugs
- Avoid concern over losing removed text
- Enables collaboration
- Documents intellectual development

Approaches

- Manual versioning
  - Use ISO dates: `2026-02-15_analysis.R`
  - Version numbers: `manuscript_v03.tex`
  - Coauthor comments: `manuscript_v03_MN.tex`
- Cloud version history (synced folders, e.g., Dropbox, Box)
- Git-based workflows (e.g., Git with GitHub)

Overview I

Writing Workflow: From Ideas to Manuscript

Linear model (inefficient):

Ideas → Notes → Data → Code → Figures → Manuscript

Key Insight: The workflow is iterative, not strictly linear.

Most time is spent revising.

Common Transition Points

- Notes → structured research questions
- Code → validated results
- Clarity → candidate figures
- Figures → narrative refinement

Reminder:

Draft sections early (especially Methods and Results).

Pre-Drafting Structural Design

Core Principle:

Design structure deliberately; diagnose structure explicitly.

Storyboarding

- Construct the argument visually before drafting prose.
- Create one slide, page, or card per figure or major claim.
- For each, specify:
  - The core claim (1 sentence)
  - The evidence shown
  - Why it matters for the overall argument
- Arrange in logical order to test narrative flow.
- Identify missing transitions or unsupported claims early.

Goal: Ensure overall logical coherence before drafting prose[4]

[4] But remain open to change!

Post-Drafting Structural Diagnosis

Core Principle:

Design structure deliberately; diagnose structure explicitly.

Reverse Outlining

- Extract 1–2 bullet-point claims per paragraph.
- Ignore prose quality; focus only on argumentative function.
- Ask:
  - Does each paragraph have a clear purpose?
  - Does the sequence of bullets form a coherent argument?
  - Are there redundancies or logical gaps?
- Revise structure before revising sentences.

Goal: Ensure both internal and overall logical coherence after drafting prose.

Overview I

Writing Tools

Word / Pages / Google Docs

Pros:

- Widely used; track-changes
- Real-time cloud collaboration

Cons:

- No or poor version control
- Formatting instability

LaTeX w/ Git, Overleaf, Prism

Pros:

- Plain text
- Superior math, refs, tables
- Auto-numbering & insertion

Cons:

- Learning curve
- Harder for collaborators
- Cloud-based track-changes/ commenting

Markdown

Pros:

- Lightweight, plain text
- Future-proof format
- Easy code integration

Cons:

- Toolchain setup required
- Limited formating & image handling

Scrivener

Pros:

- Research & reference binder
- Snapshots for safe revisions
- Great for nonlinear drafting

Cons:

- Not for collaborative work
- Proprietary file format
- Export cleanup required

Additional Tools

Dictation Tools

e.g, MS Speech Recognition, macOS Dictation, Otter.ai (300 min./mo.)

Pros:

- Rapidly capture ideas, reduce typing fatigue
- Lowers activation energy to overcome writing inertia

Cons:

- Substantial editing (typos, grammar, clarity, formatting)
- Misinterpretation of technical or domain-specific terms

Large Language Models (LLMs)

e.g., Claude, ChatGPT, Scite, Elicit, Prism, and many more

Pros:

- Assist in drafting sentences, paragraphs, and structured outlines
- Suggest structural improvements and phrasing
- Literature summaries or conceptual scaffolds

Cons:

- Risk of (over)reliance, inhibiting original thought
- Factual inaccuracies, limited domain knowledge and literature access
- Generic or formulaic writing
- Environmental impact
- Ethical concerns around authorship, bias, and misuse
- Journal restrictions

Overview I

Conceptual and Data Figures

Conceptual Figures

- Clarify theoretical frameworks
- Summarize complex relationships

Data Figures

- Present core results
- Should be interpretable independently

Manual vs. Code-Based Creation

- Manual (e.g., Illustrator): high aesthetic control
- Code-based: reproducible and scalable

File Formats Commonly Requested

- PDF – typically sufficient
- TIFF – high-resolution images
- EPS – vector graphics
- Figure resolution (e.g., 300–600 dpi)

Workflow Integration

Align structure, tools, and formatting strategy.

- Organize for clarity early to reduce friction later.
- Choose tools that match writing needs.
- Separate development and writing from formatting.

Goal:

Efficient, transparent, and reproducible writing workflow.

Overview I

Metastructure

Overview I

Audience Identification

- Know your target audience:
  - General scientific audience
  - Specialists vs. interdisciplinary researchers
  - Theoreticians and/or empiricists
- Audience shapes decisions about:
  - Terminology and background explanation
  - Context and significance framing
  - Level of technical detail
  - Visual presentation choices (e.g., conceptual figs)
- Tailor writing to match audience expertise and interests
- Consider what audience knows vs. needs to know

Journal Selection

- Identify target journals *before* you start writing
- Rank potential journals:
  - Aspirational journals (high impact, broad audience)
  - Realistic journals (good fit for topic and scope)
  - Backup journals (alternative options if rejected)
- Consider:
  - Similarity of published articles to your work
  - Journal scope and audience
  - Open access policies (and costs)
  - Impact factor and prestige (with caution)
- If you're not confident
  - Check target journal requirements for alignment
  - Review recent issues

Overview I

Signposting

- Guide readers through your argument with effective signposting
- Use transitions and connectives:
  - Topic sentences that signal direction
  - Transitional phrases linking paragraphs and sections
  - Explicit statements of relationships between ideas
- Employ structural markers:
  - Clear section headings and subheadings
  - Summary sentences to recap key points
- Help readers anticipate and understand:
  - What comes next
  - How ideas connect
  - Why information matters

Structural Coherence

- Maintain logical flow throughout the paper
- Structure provides the “roadmap” for readers
- Key aspects:
  - Logical progression of ideas within sections
  - Clear connections between sections
  - Motivation established before presenting methods
  - Results presented before interpretation
  - Conclusions grounded in findings
- Ensure each section connects to and builds on previous

Overview I

IMRaD (Introduction–Methods–Results–Discussion)

1. **Methods** first (most structurally constrained).
2. **Results** second (figure-driven; claim-evidence alignment).
3. **Introduction** after results clarify true contribution.
4. **Discussion** last to interpret, generalize, and situate.

Introduction (Why?)

- Move from broad problem to specific knowledge gap.
- Critically evaluate literature (don't summarize passively).
- End with explicit research questions, hypotheses, or key finding.

Methods (How?)

- Provide enough detail for replication.
- Use clear subheadings aligned with research questions.
- Justify key analytical decisions.

Results (What?)

- Organize around figures/tables.
- Lead each subsection with the main finding, align with Methods sections
- Avoid interpretation beyond minimal context.

Discussion (So what?)

- Interpret findings relative to hypotheses

Methods-at-the-End Papers

Design a Figure-Driven Narrative

- Determine the logical sequence of core results.
- Ensure each figure advances the central claim.
- Remove peripheral analyses from main text.

Front-Load Essential Context

- Provide minimal methodological orientation in Results.
- Define key variables and datasets before presenting outcomes.

Write Results as a Coherent Argument

- Each subsection should answer a focused question.
- Maintain conceptual momentum between figures.

Develop Detailed Methods Section at End

- Provide full reproducibility detail.
- Use clear subsections for sampling, measurements, analyses.
- Anticipate reviewer questions and address proactively.

Key: Requires exceptionally clear storytelling

Literature Reviews

- **Define scope precisely**
  - What question is the review answering?
  - What time period, disciplines, or methodologies are included?
- **Develop a transparent search strategy**
  - Databases used, search terms, inclusion/exclusion criteria.
  - Consider a flow diagram for study selection.
- **Organize by conceptual structure (not chronology)**
  - Thematic clusters
  - Competing hypotheses
  - Methodological approaches
- **Synthesize within sections**
  - Compare studies directly.
  - Highlight agreements, disagreements, and methodological drivers.
- **Conclude with implications**
  - Identify unresolved questions.
  - Clarify future research directions.

Synthesis

- **Begin with a conceptual problem**
  - What tension, paradox, or fragmentation exists?
- **Abstract upward**
  - Move from specific findings to higher-level principles.
- **Integrate across domains**
  - Combine theories, datasets, or methods not typically linked.
- **Develop a unifying framework**
  - Conceptual diagram or model.
  - Clear definitions of key constructs.
- **Demonstrate explanatory gain**
  - Show how the synthesis resolves contradictions or generates predictions.

Data Papers

- Clarify dataset purpose and potential reuse scenarios.
- Provide full documentation:
  - Sampling design, instrumentation, preprocessing steps.
  - File structure and variable-level metadata.
- Demonstrate quality control
  - Validation checks, error screening, uncertainty description.
- Include example use cases
  - Illustrative analyses showing dataset functionality.

Overview I

Methods, Results & Supplementary Materials

Methods and Results

- Sufficient detail for general understanding & reproducibility
- Balance completeness with readability
- Use parallel subheadings to organize

Supplementary Materials

- Use strategically to keep main text focused and concise
- Main text must remain complete and independent
- Include:
  - Extended methods, secondary or alternative analyses
  - Statistical analysis details or results tables
  - Details and results that are not essential to main narrative

Integration of Equations

- Introduce equations before presenting them
- Interpret equations after presenting them
- Equations (symbols) are words in a sentence, not standalone objects (Use appropriate grammar and punctuation)
- Guidelines:
  - Define all variables and parameters
  - Explain the logic and significance of equations to broad audience
  - Follow standard (or explain) mathematical notation
- Keep equations in the main text only if essential to narrative[5]

[5] See Fawcett & Higginson, *PNAS* 2012 and responding commentaries

Designing an Effective Title

Most-read part — determines if reader even looks at paper!

- **Be specific:** state the finding, not the topic
- **Aim for 8–14 words**: enough to be precise, short enough to scan
- **Front-load key terms** for visibility
  Avoid: *"A study of..."*, *"The role of..."*, *"Effects of X on Y"*
- **Consider style:** declarative sentence, question, or noun phrase
  A noun phrase has no main verb (essentially a sophisticated label). *"Temperature dependence of..."* names phenomenon without asserting anything about it.
- **Catchy, clear, concise**[6]

Weak	*A Study of predator foraging behavior in an aquatic system*
Stronger	*Temperature mediates Type II functional responses in a dragonfly–Daphnia predator–prey system*
Strongest	*Temperature dependence of attack rates but not handling times in a dragonfly–Daphnia predator–prey system*

[6] See Singh Chawla (2025)

Structuring an Effective Abstract

1–2 sentences Broad introduction to the field, accessible to any scientist

2–3 sentences Detailed background for scientists in related disciplines

1 sentence The specific problem this study addresses

1 sentence The main result — use *"here we show"* or equivalent

2–3 sentences What the result reveals relative to prior understanding; what it adds to existing knowledge

1–2 sentences Broader context and implications

Key Principles[7]

- Self-contained: interpretable without access to the full paper
- Concise: limit to essential details and key message
- Include quantitative results where possible (e.g., *"attack rates declined by 43% at 10° C relative to 20° C, with handling time showing no change."*)
- State implications explicitly: do not leave the reader to infer them

[7] See *Nature-Summary-Paragraph.pdf*

Overview I

Write Concisely

Brevity signals confidence in argument — length can signal the opposite

Principles

- **One idea per sentence;** one purpose per paragraph
- **Cut throat-clearing:** 1st sentence can often be deleted
- **Prefer the shorter word:** *"use"* not *"utilize"*; *"show"* not *"demonstrate"*
- **Avoid redundancy:** *"completely eliminate"*, *"end result"*, *"past history"*

A Useful Test

After drafting, ask: *does every sentence earn its place?* If a sentence neither advances the argument nor provides essential context, delete it.

Verbose	*"It is important to note that the results of the analysis that was conducted clearly demonstrate that..."*
Concise	*"The analysis shows that..."*

Discussion Writing: Implications and Speculation

The Discussion is Where Science Moves Forward

Stating only what you found is not enough. Readers need to know what your findings *mean* and where they lead.

A strong discussion . . .

- **Answers the question posed in the Introduction** — return explicitly to hypothesis or objective
- **Compares findings to prior work** — not just *"consistent with"* but *why* agreement or disagreement matters
- **States implications directly** — do not leave the reader to infer
- **Licenses informed speculation** — clearly flagged (*"we suggest. . . "*, *"one possibility is. . . "*) but not avoided
- **Identifies limitations honestly** without undermining the contribution
- **Opens new questions** — the best papers leave a field more curious than they found it

Watson & Crick (1953): *"It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material."* Had they omitted this implication, acceptance of the model would have been substantially delayed.

Figure-Driven Storytelling

Figures Are Not Decoration

A well-designed figure conveys structure, mechanism, or pattern that would require thousands of words to describe — and is more memorable.

- **Plan figures before writing:** if you can sketch your story as a sequence of figures, your narrative structure is sound
- **Each figure should make one clear claim** — if it requires extensive caption explanation, simplify
- **Figures and text must be mutually reinforcing,** not redundant — do not describe in prose what is already visible in the figure
- **Captions should be self-contained:** a reader should understand the figure without reading the main text
- **Order figures to mirror narrative arc** — the figure sequence is a parallel story

Use figures creatively and purposefully

Conceptual diagrams	Hypotheses, frameworks, food webs
Data figures	Results; patterns across treatments or gradients
System figures	Study site, organism, or experimental design
Synthesis figures	Meta-analytic summaries; effect size plots

Managing Jargon and Terminology

Jargon is a double-edged sword
Exists to enable precision, not to signal membership. Overuse narrows audience, obscures weak reasoning, and slows reader.

- **Define terms on first use** — even for a specialist audience
- **Use abbreviations sparingly** — only for terms used $\gtrsim 5$ times
- **Calibrate to audience** — what is jargon for general reader may be essential shorthand for specialist
- **Be especially careful in:** the Abstract (broadest readership), Introduction (context-setting), and Discussion (implications must be accessible)

Jargon-heavy	*"Spatiotemporal heterogeneity in resource availability mediates trophic cascade attenuation across productivity gradients."*
Clearer	*"Patchy resource availability weakens top-down control by predators in more productive systems."*

Overview I

Final Thoughts

- **Meta-structure is the skeleton** of your paper — it supports and organizes all content
- **Audience and journal** shape meta-structure — tailor to fit the norms and expectations of your target
- **Structural coherence** guides readers through your argument — signposting and logical flow are essential

Overview I

5 Paperanalysis

- Individual summary
- Small-group discussion I
- Small-group discussion II
- Takeaways
- Implementation

Paperanalysis

Session Objectives

(Hopefully) you analyzed a well-written paper before class.

Today we will identify:

- What makes scientific papers **readable**.
- What makes them **engaging**.
- What might make them **well-cited**.

Focus on the **writing**, not the science.

Overview I

Individual summary (5-7 minutes)

Look back at your worksheet. Write one sentence answering:

What is the single biggest (non-scientific) reason this paper was enjoyable to read?

Then highlight in the paper (or your worksheet)...

- one **writing technique**
- one **structural feature**

...that contributed to that experience.

Think about examples such as:

- Introduction structure
- Topic sentences
- Paragraph transitions
- Evidence progression

Overview I

Identify successful elements (15-20 minutes)

Form groups of 3–4. Each person explains:

1. One structural feature
2. One sign-posting technique
3. Why their paper was enjoyable to read (science-independent)

Ask each other:

- Where in the paper does this occur?
- What exactly did the authors do?
- Why do you think this was effective?

Overview I

Identify patterns across papers (10-15 minutes)

As a group, identify **3–5 patterns** across your papers.
Discuss:

- Structural similarities
- Sign-posting techniques
- Narrative flow
- Figure-text alignment
- Writing style

Key question:
Which of these features might make a paper more likely to be cited?

Add the patterns your group finds to the class whiteboard.

Overview I

Takeaways

Strong scientific writing depends on:

- clear, recognizable structure
- logical, sign-posted flow

Reading strong (and weak) papers carefully for their writing rather than their scientific content is one of the best ways to improve your own writing.

Overview I

Implementation (remaining time)

Apply these lessons to your own manuscript.

1. Choose **three techniques** from today's discussion.
2. Identify specific places in your manuscript to apply them.
3. Write down a short revision plan. For example:
   - Rewrite topic sentence of paragraph x to state main result
   - Add conceptual figure for mechanism in Discussion
   - Move the interpretation of x from Results to Discussion

Overview I

6 Rainbowparagraphs

Rainbowparagraphs

The Rainbow Paragraph Structure

A **colour-coded** tool for building and analyzing paragraph structure. Each sentence type is assigned a colour to ensure completeness and balance.

Red *Topic Sentence* Introduces the main idea of the paragraph.

Orange *Context* Provides background information.

Yellow/**Green** *Evidence / Data / Details* Specific facts, data, or steps in a process.

Blue *Explanation / Analysis* Explains why or how the evidence supports the topic sentence.

Purple *Conclusion* Restates the main idea or summarises significance.

Key rule

Every paragraph must begin with a clear **topic sentence**. You do *not* need to use every colour.

Why Rainbow Paragraphs?

The goal

Rainbow paragraphs are a *diagnostic* and *generative* tool — use them to audit existing writing *and* to plan new writing.

What the colours reveal:

- **Missing colours** signal structural gaps (e.g. no analysis, no conclusion).
- **Colour imbalance** may indicate an under-supported claim or an over-long digression.
- **Ambiguous sentences** — those that could be two colours — often need to be split or sharpened.

Good defaults to remember:

- Red and purple each appear *once*.
- Orange, yellow/green, and blue can each span multiple sentences.
- Paragraphs in *Methods* sections naturally use fewer colours.

Activity

Hopefully: you brought (1) a written paragraph from your *Introduction*, *Results*, or *Discussion*, and (2) an outline for an unwritten paragraph from either section.

Part 1 — Colour-code a classmate's paragraph

- Exchange your written paragraph with a classmate.
- Colour-code their paragraph *(up to 15 min)*, then colour-code your own *(up to 10 min)*.
- Return paragraphs; compare colour-codings and discuss differences — what made sentences easy or difficult to categorize? *(up to 15 min)*

Part 2 — Outline an unwritten paragraph

- Colour-code the outline of your unwritten paragraph and elaborate as desired *(up to 10 min)*.
- Explain the content and structure of your paragraph to your classmate *(10 min each)*.

We'll use the remaining class period to reflect on the activity as a group.

Overview I

Unfilteredflow

Unfiltered Flow: High-Velocity Drafting

Goals

- Reduce perfectionism and lower barriers to starting
- Identify core ideas by writing without premature editing

Approach

- Short, timed writing sprints
- No deleting, no revising
- Focus on momentum

`https://www.squibler.io/dangerous-writing-prompt-app`

Overview

Zero Filter

Prompt

What is the worst possible first sentence you could write for a scientific paper?

Rules

- 5 minutes
- Do not aim for quality; lean into absurdity or imperfection. Write badly!

Intention

- Reduce perfectionism and cognitive inhibition
- Prime fluency and lower the barrier to starting

Debrief

- What did it feel like to deliberately write badly?
- Did removing quality expectations change your writing speed or tone?

Stream of Consciousness

Prompt

Write everything you already know about your research topic—no structure.

Rules

- 5 minutes
- No deleting or revising
- If stuck, repeat the last word until a new idea emerges

Intention

- Externalize tacit knowledge
- Reveal gaps or redundancies in understanding

Debrief

- What surprised you about what you do (or do not) know?
- Did any new connections emerge?

Overview I

Jargon Free

Prompt

Explain your research question without using any technical jargon.

Rules

- 8 minutes
- Avoid field-specific terminology entirely
- If you accidentally use jargon, immediately rephrase and continue

Intention

- Improve conceptual clarity and accessibility
- Strengthen introductions and broader-impact framing

Debrief

- Which concepts were hardest to simplify?
- How might this improve your manuscript's readability?

Volume Competition

Prompt

Write as much as possible describing your study system or methods.

Rules

- 8 minutes
- Word count determines the winner
- Content must remain at least minimally coherent

Intention

- Maintain baseline coherence under pressure

Debrief

- Did speed compromise clarity?
- What strategies helped you maintain momentum?

Overview I

Reverse Logic

Prompt

Argue why your central hypothesis is wrong.

Rules

- 10 minutes
- Construct the strongest possible counterargument

Intention

- Sharpen critical thinking
- Anticipate reviewer critiques

Debrief

- Did this reveal any real weaknesses?
- How could you preempt these critiques in your manuscript?

Methods in Motion

Prompt

Describe your methods section as if explaining it to a colleague who wants to replicate your work.

Rules

- 10 minutes
- Focus on sequence and logic, not polish
- Keep typing even if details feel incomplete

Intention

- Translate procedural knowledge into structured prose
- Identify missing steps or ambiguities

Debrief

- Where did you hesitate or lack detail?
- What parts felt most clear vs. underdeveloped?

Figure Narrative

Prompt

Write the results section for one key figure in your manuscript.

Rules

- 12 minutes
- Describe patterns, trends, and interpretations
- Do not reference the figure explicitly; make the text stand alone

Intention

- Strengthen the link between data and narrative interpretation

Debrief

- Did the writing clearly convey the main result?
- What additional analysis or visualization might strengthen this section?

Abstract Compression

Prompt

Draft a complete abstract for your manuscript.

Rules

- 12 minutes
- Must include context, question, methods, key results, and implications
- Prioritize completeness over perfection

Intention

- Synthesize a coherent high-level narrative

Debrief

- Which component was hardest to articulate quickly?
- How does this draft compare to your current abstract (if one exists)?

Overview I

Closing Reflection

- Which activity generated the strongest usable text?
- Which activity exposed the biggest conceptual or structural gap?
- What one unfiltered-flow practice will you carry into your weekly writing routine?

Overview I

Reverseoutline

What Is a Reverse Outline?

Instead of planning what you want to say, map what you actually said.

Diagnose structural problems

Redundancy, misplacement, missing links, and logical gaps.

Reverse outlining reveals paragraphs that:

- drift from the intended argument
- mix multiple ideas that should be separated
- do not clearly support the central claim

The Process

1. Analyze your manuscript paragraph by paragraph.
2. Write a short summary of each paragraph's **purpose**.
3. Use summaries as a **structural map** of the manuscript.

For each paragraph, answer two core questions:

1) What it says (*Topic*)

The main idea or point being made.

2) What it does (*Function*)

How the paragraph serves the manuscript, its argument, and your audience.

Activity: Reverse Outline Your Paragraphs

Part 1 (up to 20 minutes)

- Reverse outline each paragraph as objectively as possible.
- Do not edit yet, just take notes.

Part 2 (up to 10 minutes)

- Share your paragraphs with your neighbour.
- Reverse outline each others' paragraphs.

Part 3 (up to 15 minutes)

- Use both reverse outlines to assess whether each paragraph's sentences say what they need to say and do.

Part 4 (up to 15 minutes)

- Shift to a manuscript-level perspective.
- Assess how each paragraph, as a whole, serves the manuscript and its audience.

We'll use the remaining class period to reflect as a group or continue reverse outlining additional paragraphs.

Overview I

Literature

Overview I

The Literature Problem

Two Competing Demands

- Knowing the **foundational and classic** literature in your field
- Staying current with a **rapidly expanding** body of new work — often across multiple subfields

Common Failure Modes

- **Passive accumulation:** PDF hoarding without reading or annotation
- **Over-reading without synthesis:** consuming papers without integrating them into an argument
- **Reliance on incomplete system:** searching only Google Scholar, relying only on specific keyword alert

Key Principle

You do not need to read everything. You need a **structured system** for filtering, organising, and revisiting literature that balances skimming and deep reading.

Overview I

Staying Current: Alerts and Feeds

Set up alerts — then let new literature come to you

Journal ToC / bioRχv Alerts Subscribe through websites or aggregate via RSS reader (e.g., Feedly). Best for core journals in your subfield — gives you everything, not just keyword matches.

Web of Science / Google Scholar Alerts Key paper and keyword alerts delivered by email on a schedule you control. *Highly filtered; low noise.*

ResearchGate / Google Scholar Profile Suggestions Follow authors in your field, surface papers based on your reading history.

Social Media Direct announcements of new papers; hashtags and community accounts surface work quickly. *Informal but fast.*

Strategy

Separate **scanning** from **deep reading**; schedule dedicated time for each.

Staying Current: Active Filtering

Three-Tier Filtering System

1. **Scan:** titles / abstracts only — decide relevance in under 60 sec.
2. **Triage:** save potentially relevant PDFs to reference manager or folder
3. **Deep Read:** full reading reserved for highest-priority papers

Heuristics for Prioritisation

- Direct relevance to current or future projects
- Methods, datasets, or frameworks of potential use
- Intriguing novelty / fun

AI-Assisted Reading Tools (examples)

NotebookLM	Query PDFs conversationally; rapid orientation to new subfield
Speechify	Convert papers to audio for assessment in low-focus time

Reading everything is inefficient

Selective re-reading is where understanding develops.

Overview I

Literature Search Strategies (Traditional)

Citation Chaining

- **Backward search:** scan for common references in recent papers
- **Forward search:** scan and follow citations of key papers

Author Search

- Identify 5–10 leading (frequently cited) researchers; track the citation records of most cited papers

Start Broad, Then Narrow

- Use general keywords to identify core papers; refine iteratively using Boolean operators (`AND`, `OR`, `NOT`)
- Update as terminology, useful keywords and authors become clearer

Goal (especially for new-to-you fields)

Identify the **core intellectual structure** of a topic, not just a list of papers.

Literature Search Tools

Web of Science Curated index; strong citation tracking and bibliometric tools; advanced Boolean searches and exportable results. The Standard for systematic searches. *Requires institutional access.*

Google Scholar Far less precise than WoS but broad coverage including preprints, theses, and grey literature. Useful for quick searches and obtaining PDFs. *Free.*

AI-Assisted Tools Tools evolving rapidly. Reliability growing but TBD.

- **Elicit:** structured queries and paper summaries
- **Scite:** identifies whether papers *support* or *contrast* the work they cite
- **Semantic Scholar / ResearchRabbit:** relevance ranking, topic mapping, and citation network visualisation

Overview I

Reference Managers

Core Function

Database to organize sources and auto-format citations and bibliographies

Other Functions

- Enable tagging, search, and annotation
- Store and organise PDFs
- Sync across devices; shared group libraries

Zotero[8] (most used?)

- Integrates with Word, Google, LaTeX (w/ `Better BibTeX`)
- Browser plugin captures metadata and PDFs in one click
- Group libraries: share collections with co-authors in real time
- Supports tags, notes, related items, PDF annotation

Alternatives include: BibDesk[1] (LaTeX), Mendeley[2] (strong PDF reader), EndNote[2] (institutional standard), Paperpile[2] (Google Docs integration)

[8] Free (maybe limited) and open-source; [2] **$$$** proprietary

Organizing Sources

Build Structure, Not Just Storage

- Organise by **project** or **manuscript**, not by topic alone — projects have deliverables
- Use **tags** for cross-cutting themes (e.g., `classic`, `2read`, `cited-in-ch2`)

Recommended Approach: Two-Layer System

- **Reference manager:** storage, PDF access, citation generation
- **External notes** (e.g., a plain-text or Markdown file per project): conceptual synthesis, argument mapping, connections across papers

The two layers complement each other

The reference manager stores, the notes file thinks

Annotated Bibliographies

Purpose

- Convert reading into structured, reusable knowledge
- Force active engagement (you can't annotate what you've only skimmed)
- Becomes raw material for your Introduction and Discussion sections

What to Include per Entry

- Core question or objective of the paper
- Key findings and conclusions
- Methods or analytical approach
- Relevance to your own work
- Limitations or unresolved gaps

Suggestions

- Write in your own words — not copy-pasted from abstract
- Keep entries concise: 3–6 bullet points
- Use Zotero's note field to keep with reference
- Update entries as your understanding evolves

Overview I

Integrating Literature in Your Writing

Writing with specific papers in mind

- *Pros:* precise citation; accurate representation of prior work; anchors claims to evidence
- *Cons:* can produce fragmented, citation-heavy prose; limits synthesis; writing follows the literature rather than your argument
- *Best for:* methods sections, direct comparisons, specific statistics

Writing without specific papers in mind

- *Pros:* promotes conceptual clarity; argument-driven structure; encourages synthesis, faster drafting[9]
- *Cons:* risk of vague/unsupportable claims; requires well-stocked library; misrepresented citations
- *Best for:* introduction and discussion (synthesis)

[9] *"Write drunk, edit sober."* — Hemingway

Best Practice: The Hybrid Writing Strategy

Step 1 — Draft Conceptually

- Write the argument without citations
- Focus on logic, structure, and your own voice
- Placeholder (`[CITE]`) where citation is needed

Step 2 — Map Literature Onto the Argument

- Add citations to support each claim
- Replace general statements with specific evidence
- Consult annotated bibliography — most sources are likely there

Step 3 — Refine for Balance

- Avoid over-citation — not every sentence needs a reference
- Ensure key and foundational works are represented
- Verify every citation against the source
- Never cite a paper you have not fully read

Recommendation

Use argument-first writing for structure and flow, then verify every citation.

Takeaway

Knowing the classics and staying on top of the literature is a **process** and is a part of your **job** as a scientist (and educator).

Recognize it as such and **regularize it** as part of your weekly schedule.

Remember the four pillars:

- **Discover:** alerts + citation chaining + active search
- **Filter:** scan → triage → deep read
- **Organise:** reference manager + external synthesis notes + annotated bibliography
- **Integrate:** argument-first drafting → citation mapping → refinement

Takeaway

Managing the literature is not about volume.
It's about **structure, selectivity, and synthesis**.

Overview I

Collaboration

Overview I

Dissertation Manuscripts as Collaborative Efforts

- **You own the draft.** As first author, you are responsible for maintaining *momentum* on the manuscript and for ensuring its *integrity*.
- **Advisors may be co-authors, don't expect them to be editors.** They may choose to revise and redirect, but the *intellectual initiative* should come from you.
- **Co-authorship is not automatic.** Contribution, not relationship or seniority, should determine authorship; clarify expectations *before* writing begins.
- **Co-authors have competing priorities.** You, your advisor, and your collaborators need not share the same *incentives*.

Overview I

Tools for Collaborative Writing

Choose tools that match your collaboration needs

MS Word / Apple Pages Share via email or cloud storage. Well-known; track changes; offline editing; weak for complex or equation-heavy formatting.

Google Docs Real-time cloud-based editing. Low barrier to entry; suggesting mode; comment threads; best for early drafts or non-LaTeX collaborators.

Overleaf Cloud-based LaTeX editor with real-time collaboration. Comment threads; full revision history; Git sync; the standard for LaTeX manuscripts with multiple authors.

Git / GitHub Version-controlled repositories; cloud-based. Full audit trail; branching; ideal for LaTeX and code; not real-time; steeper learning curve.

Practical Rule: Agree on a tool *before* writing begins. Mid-project migrations are time-consuming and introduce errors.

File Versioning

If you're not using version control software (Git)...

File Naming Ambiguous filenames are a form of technical debt. Adopt a convention and enforce it from day one.

Avoid	`draft_final_v2_JSmith_revised_FINAL.doc`
Better	`2025-11-14_manuscript.doc`
Better	`manuscript_v07.doc`

Versioning Conventions

- Use **ISO dates** (`YYYY-MM-DD`) or **sequential version numbers** (`v01`, `v02`) — not both. Append commenting author initials.
- **Tag submission-ready versions** explicitly (`v_submitted`, `v_R1`)
- **Don't delete old versions** — archive them in an `/archive` subfolder

Comment and Annotation Discipline

- Resolve or explicitly defer all comments before circulating a new draft
- Use GitHub Issues or margin note comment threads to track unresolved questions and decisions
- Do not leave unresolved comments at submission — some journal portals expose tracked changes to reviewers

Overview I

Establishing Norms and Expectations

Collaboration is Communication

Most conflicts stem from unmet unstated expectations

Clarify at the start:

- **Authorship:** Confirm whether your advisor will be last author and who is to be offered (the right to decline) authorship.
- **Level of involvement:** Will advisor/collaborators co-write sections, provide key figures, or provide feedback on drafts?
- **Escalation path:** Who has final say on journal choice, framing, interpretation, etc.? What happens if co-authors disagree?
- **Primary channel:** email, Slack, GitHub Issues — consider one platform for decisions, one for quick questions.
- **Response expectations:** what is a reasonable turnaround for draft review (1 week? 3 weeks?) and for quick questions (48 hours?).
- **Meeting cadence:** regular check-ins vs. milestone-driven meetings — both have costs and benefits.

Decision documentation: For large collaborations; all substantive decisions recorded in a shared log (email thread, shared doc, etc.)

Common Issues

Common Friction Points

Advisor rewrites your draft: Treat as mentorship; study the changes; ask for rationale on substantive edits.

Co-author delays for weeks: Set a gentle deadline when providing updated draft — *"Could you get it back to me by X."*

Co-authors disagree: Raise the conflict openly; do not triangulate between them; request a joint discussion.

A Common Failure Mode

Decisions made verbally in hallway conversations or lab meetings that are not recorded. Three months later, co-authors remember different outcomes.

Overview I

Resolving Disagreements

Identify the Type of Disagreement First

Type	Example	Possible Resolution Pathways
Scientific	Disagreement over statistical model choice or interpretation of results	Resolve with evidence; consider additional analyses; consult a neutral expert
Framing	Disagreement over how to position paper's contribution	Consult target journal scope; summarize framing of comparable studies
Stylistic	Disagreement over sentence-level writing choices	Defer to the lead author; style is not worth extended negotiation
Interpersonal	Conflict rooted in prior relationship dynamics, credit concerns, or workload resentment	Address directly and early; do not let it contaminate scientific decisions

- Separate *positions* (what someone wants) from *interests* (why they want it) — interests are usually reconcilable
- Put manuscript's quality and reader's needs at the center of dispute
- Seek advice from neutral party (e.g., committee member) if direct resolution is failing

Overview I

Dividing and Tracking Contributions

Ambiguity about responsibility is a project risk

Task Delegation Principles

- **Assign ownership, not just involvement** — one person is responsible for each deliverable, even if others contribute
- **Be explicit about scope:** *"draft the methods section"* is a task; *"help with methods"* is not
- **Revisit assignments** as the project evolves — initial task maps rarely survive contact with data
- **Distinguish writing tasks from research tasks:** the person who ran the analysis may not be the best person to write it up, and vice versa

Contribution Tracking Tools

Google Sheets Simple shared task list with owner, status, and deadline columns — low barrier, universally accessible.

GitHub Issues Create tasks with milestone tracking; integrates directly with version-controlled repositories.

Asana/Trello/Notion Visual project boards suited to larger or longer-running collaborations.

Overview I

The Authorship Conversation

When to Have It

Early — ideally before data collection begins, certainly before writing starts. Revisit at submission. (Conversations become exponentially harder once significant effort has been invested.)

What to Establish

- Who will be an author vs. acknowledged contributor?
- What is the expected authorship order and on what basis?
- What happens if contributions change substantially during project?
- Who is the senior and the corresponding author?
- What is the plan if a co-author becomes unresponsive before submission?

Avoid

- **Gift authorship:** including advisor/collaborator who did not contribute substantially/intellectually
- **Ghost authorship:** omitting those who contributied substantially
- **Retroactive negotiation:** changing authorship order after submission without full co-author agreement

Overview I

Hierarchy and Power Imbalance

Common Sources of Imbalance

- **Career stage:** harder for student to push back on advisor's framing choices, even when student is domain expert
- **Disciplinary norms:** fields may have incompatible assumptions about authorship order, target journal status, publication timeline
- **Publication pressure:** misaligned incentives around when and where to publish can fracture teams

Power Differentials Are Structural, Not Personal

Collaborations can involve genuine differences in career vulnerability and publication pressure. Ignoring these dynamics does not neutralize them.

Protective Strategies

- Document your contributions in writing as work proceeds — do not rely on memory or goodwill at submission time
- Raise authorship expectations explicitly and early, before significant work has been invested
- Consider drafting the *Authorship Contributions* statement early

Overview I

Final Thoughts

Collaborative writing is a skill that is eased by intentionality.

Key Takeaways

- Establish norms explicitly and early
- Communicate openly about expectations and contributions
- Address conflicts directly and with manuscript quality in mind
- Document major decisions and contributions openly

Overview I

Peerreview

Overview I

The Peer Review Pipeline

What happens after you hit submit?

1. **Submission** Manuscript enters the journal's editorial system; author receives an acknowledgement and manuscript ID.
2. **Editorial Triage** Editor-in-Chief or handling editor assesses scope fit and minimum quality threshold. Desk rejection occurs often within days without external review.
3. **Reviewer Invitation** Editor invites 2–4 reviewers; many (most) decline. This stage can take days to several weeks.
4. **External Review** Reviewers evaluate the manuscript and return written reports, within 2–8 weeks (often longer in practice).
5. **Editorial Decision** Editor synthesises reviews into one of four (or more) outcomes: **accept**, **minor revision**, **major revision**, or **reject** (with or without prejudice).
6. **Revision & Resubmission** Authors revise and submit response letter.
7. **Acceptance & Production** Manuscript enters copyediting, typesetting, and proofing before online publication.

Typical timeline: first decision in 3–5 months; full acceptance in 6–18 months across all revision rounds.

Overview I

The Editor's Perspective

What editors are looking at

- **Scope:** does the manuscript fit the journal's aims and readership?
- **Significance:** does it advance the field enough to justify publication?
- **Novelty:** is it sufficiently distinct from recent literature?
- **Presentation:** is the manuscript professionally prepared?

Why desk rejection happens

- Obvious scope mismatch
- Insufficient (perceived) methodological rigour on first read
- Incomplete submissions (missing figures, references, etc.)
- Manuscript prepared for a different journal (wrong format, wrong citation style)

What this means for you as an author

- Tailor your cover letter to the journal
- Polished submissions reduces editorial hesitation
- Desk rejection is typically not a judgement of your science — it is usually a targeting problem

Overview I

The Reviewer's Perspective

What reviewers evaluate

- Conceptual framing, novelty, significance of research question
- Appropriateness and rigour of methods
- Validity of results and statistical analyses
- Interpretation: do conclusions follow from the data?
- Presentation: clarity, figures, writing quality
- Completeness of citations and engagement with prior work

What reviewers shouldn't be required to do

- Editing your manuscript for you
- Reading a poorly organised paper charitably

Reviewers are busy volunteers

- Clear, well-structured manuscripts reduce reviewer burden and tend to produce more constructive feedback
- Reviewer fatigue is real: dense, difficult manuscripts are evaluated less generously

Tip: as you revise your writing, ask yourself — *am I making it easy for a tired expert to understand and trust this work?*

Overview I

The Author's Perspective

Accept Rare on first submission. Minor corrections likely requested at proofing stage.

Minor Revision The manuscript is broadly acceptable; targeted changes required. Typical turnaround: 2–4 weeks.

Major Revision Substantial work required (new analyses, restructuring, additional experiments). Usually one more full review round. Typical turnaround: 4–12 weeks. *This is the most common outcome.*

Reject The manuscript will not be reconsidered at this journal.[10] May include encouragement to resubmit elsewhere.

[10] Entirely new manuscript expected for "rejection without prejudice."

First steps

First steps after receiving reviews

1. Read once for overall tone — then **step away**
2. Return the next day and read for content, not emotion
3. Identify which comments require new analysis, which require writing, and which require clarification only
4. Distribute your synthesis and discuss a plan with co-authors *before* revising

Rejection Management

Rejection is normal — not exceptional

- Rates can exceed 70–90%; desk rejection of 50%+ at high-impact journals
- Most published papers were rejected at least once
- Rejection does not mean the work is unpublishable — it usually means the targeting was wrong

What to do immediately after rejection

- Read the decision letter and reviews carefully — even a rejection often contains actionable feedback
- Discuss with co-authors: is the feedback *scientific* (revise before resubmitting) or *editorial* (wrong journal)?
- Identify the next target journal *before* making any revisions

Appealing: appropriate if review contains clear and foundational factual error or demonstrable misreading of the work. Rarely successful, can take weeks or months, and delays resubmission elsewhere

Rule of thumb: Rejection sucks. Get over it and aim to submit a rejected manuscript elsewhere quickly ($<$ 4–6 weeks). **Momentum matters!**

Overview I

Revision & Resubmission Workflow

Organise before you write

- Extract all editor and reviewer comments into a single numbered list
- Identify overlap and dependencies:
  - Overlap across reviewers' comments
  - Revisions requiring new analyses prior to writing
- Set a revision timeline (considering journal's request); request extension early if needed

Revise manuscript and draft response letter in parallel

- Draft response as each comment is addressed, not retrospectively
- Track changes in manuscript (or use LaTeX diff tool)
- Keep change-tracked version for editor

Verify

- Every response references a line number in the revision
- Every comment has a response (even just "*Implemented on line x.*")
- Changes haven't broken figure or table references

Responding to Reviewer Feedback

The response document is as important as the revision

- Editors often read your response *before* re-reading the manuscript — make it clear and complete
- Address **every** comment, even if you disagree
- Use a structured format: quote the comment, state your response, indicate the change made and its location[11]

Handle disagreement professionally

- You may respectfully decline a reviewer request — but you must explain why clearly and with evidence
- Phrase pushback as: *"We respectfully disagree because... and instead propose..."*
- Never argue with tone; argue with data and logic
- If two reviewers contradict each other, flag it to the editor, use it to your advantage (if appropriate), and propose a resolution

[11] See provided LaTeX template.

Overview I

The Proofing Process

Proofing is a **correction stage**, not a revision stage.
Increasingly done through journal's online proofing system.

Read very carefully, especially...

- All author names, affiliations, and corresponding author details
- All equation, special character, and Greek symbol rendering
- Figures and tables in the correct order, at sufficient resolution, correctly captioned

Practical notes

- Proofing deadlines are typically 48–72 hours
- Read proof against manuscript line by line, not from memory
- Uncaught errors become permanent in published record; corrections require formal erratum

Remember to celebrate!

Overview I

Peer Review History

What did you learn from reading the peer review history of published papers?

For discussion:

- What stood out in the reviewers' comments?
- What stood out in the editor's decision letters?
- What did the authors do well in their responses?
- What would you have done differently?

Overview I

12 Peerintroanalysis

- Format
- Feedback Questions
- Debrief

Peerintroanalysis

Overview I

Analysis of Peer Introductions

Group format

1. Work in groups of three.
2. Read each peer's introduction draft (∼5-10 min each)
3. Then, for each of your two peers, provide written feedback (∼15 min). (*Questions on next slide.*)
4. After everyone is finished, explain your answers to each peer (∼10 min).

Focus of this activity

We are not evaluating scientific correctness. We are evaluating clarity, structure, and audience engagement.

Overview I

Feedback Questions

For each peer's introduction, answer:

1. What did the author do well?
2. Was the topic introduced in a compelling way? Was the intended audience inferrable and was the topic cast at the right level? Suggest a means by which it might it be reworked to be more compelling and/or engage a wider audience.
3. Paraphrase the manuscript's specific question in 1–3 sentences. Was it clear?
4. Summarize the logicical flow in 4–8 words (e.g., x $\rightarrow$ y $\rightarrow$ z). Was the logical flow clear? Identify one place that could be clarified.
5. If the Introduction ended with a take-home message, what was it? Paraphrase it in 1–3 sentences. Assess whether it matched your expectations created by the rest of the Introduction.

Overview I

Debrief and Reflection

- What were common suggestions?
- What did you learn about your own writing?
- What characterized helpful feedback?

Overview I

Journalprep

Overview I

Author Guidelines

Read them! They can be overwhelming, but focus on the essentials.

Things I ignore:

- Font type and size, margin requirements
- Journal-requested citation format

Things I don't ignore:

- Word/page limits (*incl. title, abstract, main text*)
- Line spacing (*double*) & page numbering
- Figure placement (*inline unless specified*) and formats (*pdf, tiff, eps*)
- Cover page and/or end-of-manuscript requirements
  - Author Contribution Statement
  - Open Science (Data & Code Availability) Statement

Check:

- Journal's double-blind requirements
- Citation completeness and consistency
- Triple-read for spelling and punctuation

Overview I

Author Contributions Statement

Purpose

- Distinguish **authorship** from **acknowledgment**
- Increase transparency and accountability
- Expose *ghost authorship* and *gift authorship*
- Increasingly indexed by databases (e.g., OpenAlex)

The CRediT Taxonomy (14 Roles) `credit.niso.org`

Conceptualization	Methodology
Data Curation	Project Administration
Formal Analysis	Resources
Funding Acquisition	Software
Investigation	Supervision
Validation	Visualization
Writing — Original Draft	Writing — Review & Editing

Author Contributions Statement

Common Formats

1. **Prose (narrative)**
   "J.S. conceived the study and led data collection. M.R. performed statistical analysis..."

2. **Role-tagged list**
   Conceptualization: J.S., M.R.
   Formal Analysis: M.R.
   Writing–Original Draft: J.S.
   Writing–Review & Editing: All authors

Imperative: Agree on (a draft) contributions statement *before writing starts*, not retroactively. Disagreements are uncomfortable, potentially costly, and typically avoidable.[12]

[12] See *Collaboration* class.

Overview I

Page Charges & Publication Fees

- **Article Processing Charges (APCs):** \$500–\$12,000+ depending on journal/Open Access (OA) model
- **Page charges:** \$0–\$200 per printed page for subscription journals
- **Color figure fees:** \$300–\$1,000+ per figure; often free online-only

Who Pays — and How to Find Out

- **Ask your advisor first:** most PIs budget for publication costs through grants; this is a normal conversation to have
- **Institutional waivers:** many universities (but not OSU!) have OA publishing funds; check with library
- **Funder mandates:** NIH, Wellcome Trust, and others *require* OA and may cover fees directly
- **Request a waiver:** most journals offer hardship waivers for authors without funding — ask before assuming you must pay

Bottom line: You shouldn't be expected to pay out of pocket. Clarify funding *before* submission, not after acceptance.

Overview I

Pre-Submission Review by Friendly Reviewers

Selecting Reviewers

- Avoid/Seek reviewers with strong prior positions on your findings
- Prioritize people who will give *honest* feedback, not reassurance
- Consider who might make a better journal reviewer

Briefing Reviewers

- State the target journal and audience
- Set a realistic deadline (2-3 weeks is standard)
- Clarify what kind of feedback is useful: structure, argument, clarity, or all three
- Identify specific concerns you want addressed

Thank them by name in the Acknowledgments.
Consider mentioning in Cover Letter.
Friendly review is *not* a substitute for rigorous self-revision. Send out only when you believe the manuscript is essentially complete.

Overview I

Suggesting Potential Reviewers

How to Identify

- Researchers who publish directly on topic or methods
- Authors cited in manuscript (but not exclusively)
- Individuals with complementary expertise (theory, methods, system)

Who to Avoid

- Current or recent collaborators (typically last 3–5 years)
- Members of your institution or close professional network
- Advisors, committee members, or mentors

Strategic Considerations

- Aim for 4–8 suggested reviewers
- Ensure diversity (perspectives, institutions, career stage, gender)
- Avoid suggesting only highly favorable reviewers

Opposed Reviewers

- Allowed by some journals but use very sparingly and justify![13]

[13] Consider use as *"friendly" reviewer*

Overview I

Submission Logistics

Explore your target journal's submission system!

Overview I

Open Access Models

Type	Who Pays	Description
Gold	Author / funder	Published OA[14] immediately by the journal; APC[15]
Green	Nobody	Author self-archives a preprint or accepted manuscript in a repository (e.g., biorχiv)
Diamond	Nobody	Community- or institution-funded; no fees for authors or readers
Hybrid	Author	Subscription journal offers OA for individual articles upon APC payment
Bronze	Nobody	Freely readable on the publisher site *without* open licence; access can be revoked

Note: Green OA costs nothing and is compatible with most journals. Check policies at `https://openpolicyfinder.jisc.ac.uk`.

[14] OA - Open Access
[15] APC - Article Processing Charges (can be substantial)

Data & Code Sharing

- Increasingly required by journals and funders
- Can be cited - Digitial Object Identifier (DOI)
- Common repositories:
  - Data: Zenodo, FigShare, Data One
  - Code: GitHub, Zenodo
  - Supplementary materials (*no longer considered enough*)
- Best practices:
  - Include comprehensive documentation and README files
  - Provide appropriate licensing

Take *IB 516*: Analytical Workflows!

FAIR Data Principles

FAIR defines standards to maximise reuse and reproducibility.

F — Findable Data are assigned persistent identifiers (e.g., DOIs) and described with rich, searchable metadata.

A — Accessible Data and metadata are retrievable via open, standardised protocols; access conditions are clearly stated.

I — Interoperable Data use common formats, vocabularies, and ontologies to enable integration with other datasets and tools.

R — Reusable Data carry explicit licences (e.g., CC BY) and sufficient provenance information for others to build on them.

Full guidelines at `go-fair.org/fair-principles`.

Note: FAIR does not require data to be *open* — sensitive data can be FAIR with controlled access.

Overview I

Preprint Repositories

Consider posting your ms to a preprint server before or during journal submission.

Advantages

- **Speed:** findings are public immediately, months before formal publication
- **Priority:** timestamped record of contribution
- **Visibility:** indexed by Google Scholar, etc.
- **Feedback:** community comments before peer review
- **Open access:** freely available
- **Funder compliance:** satisfies some open science mandates immediately

Disadvantages

- **Journal policies:** some journals discourage or prohibit
- **No peer review:** errors and weak claims are public
- **Perception:** some fields and hiring committees do not credit preprints
- **Retraction gap:** flawed preprints may persist even after journal rejection
- **Media risk:** premature press coverage before validation

Preprint Repositories

Examples:

bioRχiv Life sciences; operated by Cold Spring Harbor Laboratory; the dominant biology preprint server since 2013.

EcoEvoRχiv Ecology and evolutionary biology; community-governed; supports registered reports and post-publication versions.

arχiv Physics, mathematics, computer science, quantitative biology; the oldest major preprint server (est. 1991); moderated by field endorsers.

Zenodo General-purpose; accepts preprints, datasets, code, and presentations; assigns DOIs; widely used for open science compliance.

Authorea Collaborative writing and preprint platform; supports live figures, data embedding, and inline commenting; blurs the boundary between manuscript and publication.

Peer Community In... (PCI)

`PeerCommunityIn.org`

Goals

- Transparent peer review *before* (or *instead of*) journal submission
- Eliminate fees for authors and readers
- Reduce dependence on commercial publishers

How It Works

- Authors submit preprint (e.g., to bioRχiv), then request PCI review
- Volunteer recommender (acting editor) solicits and coordinates reviews
- Accepted preprints receive a **PCI Recommendation** — citable endorsement
- PCI-friendly journals accept recommended preprints, or regular review

PCI Animal Science
PCI Archaeology
PCI Ecology
PCI Ecotox Env Chem
PCI Evol Biol
PCI Forest Wood Sci
PCI Genomics
PCI Health Mov Sci
PCI Infections
PCI Math Comp Biol
PCI Microbiology
PCI Network Science
PCI Neuroscience
PCI Nutrition
PCI Org Studies
PCI Paleontology
PCI Plants
PCI Psychology
PCI Registered Reports
PCI Statistics and Machine Learning
PCI Zoology

Take-Home Messages

- Final formatting is important, but don't obsess over minor details
- Author contributions should be agreed upon early and transparently
- Clarify publication fees and funding sources before submission
- Pre-submission review by trusted colleagues can improve your manuscript
- Suggesting appropriate reviewers can increase chances of acceptance
- Open science practices enhance transparency and impact, but consider costs and requirements
- Preprint repositories offer rapid dissemination but check journal policies